



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

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Mark BradshawMichael Drake, Joseph Pacelli, Brady Twedt

To cite this article:

Mark BradshawMichael Drake, Joseph Pacelli, Brady Twedt (2023) Brokerage House Initial Public Offerings and Analyst Forecast Quality. Management Science 69(11):7079-7094. <https://doi.org/10.1287/mnsc.2022.4610>

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Brokerage House Initial Public Offerings and Analyst Forecast Quality

Mark Bradshaw^{a,*} Michael Drake,^b Joseph Pacelli,^c Brady Twedt^d

^aCarroll School of Management, Boston College, Chestnut Hill, Massachusetts 02467; ^bSchool of Accountancy, Marriott School of Business, Brigham Young University (BYU), Provo, Utah 84604; ^cAccounting and Management Unit, Harvard Business School, Boston, Massachusetts 02163; ^dLundquist College of Business, University of Oregon, Eugene, Oregon 97408

*Corresponding author

Contact: mark.bradshaw@bc.edu,  <https://orcid.org/0000-0002-6744-8147> (MB); mikedrake@byu.edu,  <https://orcid.org/0000-0001-8112-5793> (MD); jpacelli@hbs.edu,  <https://orcid.org/0000-0003-3139-712X> (JP); btwedt@tamu.edu,  <https://orcid.org/0000-0002-7942-4249> (BT)

Received: January 11, 2021

Revised: September 9, 2021, March 3, 2022

Accepted: March 20, 2022

Published Online in Articles in Advance:
December 7, 2022

<https://doi.org/10.1287/mnsc.2022.4610>

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Abstract. We examine how brokerage firm initial public offerings (IPOs) influence the research quality of sell-side analysts employed by the brokerage. Our main results focus on earnings forecast bias and absolute forecast errors as proxies for research quality. Using a staggered difference-in-differences analysis, we document significant decreases in forecast bias and absolute forecast error during the two-year period centered on the analysts' brokerage house IPO. In additional analyses, we explore several potential explanations for the short-term benefits of brokerage house IPOs. We find some evidence that IPOs delay the departure of more talented analysts and that the effects are more concentrated among analysts and brokers that face more scrutiny.

History: Accepted by Brian Bushee, accounting.

Supplemental Material: Data are available at <https://doi.org/10.1287/mnsc.2022.4610>.

Keywords: initial public offering • financial institutions • investment • sell-side analysts • forecasting • research quality

1. Introduction

The initial public offering (IPO) is an important event in the lifecycle of any firm because it generates substantial injections of capital, facilitates the expansion of operations, and creates a public market for owners to liquidate their interests (Ritter and Welch 2002). In the financial services sector in particular, the capital raised by the IPO can have important implications for the quality of products that financial institutions provide, as well as the human capital they are able to attract and retain. Over the past few decades, there has been significant growth in IPOs in the financial sector, with this sector leading the entire IPO market in terms of deal volume (approximately \$500 billion in deal volume from 1990–2019).¹ The objective of our study is to examine the effects of IPOs on brokerage houses within the financial sector (henceforth, broker IPOs) on the quality of one of its key products—sell-side analysts' research.

We specifically focus on broker IPOs and sell-side analyst forecast quality (bias and accuracy) because of the rich academic literature examining the properties of these forecasts.² Many of these studies are motivated by conflicts of interest tied to investment banking activities (e.g., Lin and McNichols 1998, Clement 1999, Jackson 2005). Although such studies focus on the relation between the capital-raising activities of broker clients and the forecast quality of the analysts following those

clients, we were unable to find research examining how the capital-raising activities of the analysts' *own employer* might affect the quality of their research.

Studying the effects of broker IPOs on analysts' forecast quality is important because the IPO process has the potential to significantly improve analysts' ability to conduct high-quality research. First, the brokerage house may invest more in research activities during this period in anticipation of the impending infusion of IPO capital. The increase in resources available could foster higher-quality research. Second, the IPO process may lead to changes in the composition of analysts working at the brokerage house. For example, superior employees with outside options may turn over at a lower rate in expectation of "cashing in" on the IPO. Finally, it is also possible that forecast quality improves in the presence of increased scrutiny that arises during the IPO process from monitors, such as analysts, lawyers, underwriters, auditors, boards, and the business press. Such an effect would be consistent with Ball and Shivakumar (2008), who find that increased monitoring is associated with improved corporate financial reporting quality and also consistent with a large body of research across various disciplines that finds individuals are more likely to behave appropriately when they are more closely monitored (see, e.g., Bouchet et al. 1996, Kedia and Rajgopal 2011).

We note, however, that an increase in forecast quality is not obvious *ex ante*. IPOs create significant economic incentives for firms to report profitability and growth to shareholders. Consistent with this argument, prior research finds that IPO firms are more likely to take actions to boost short-term earnings after going public (Teoh et al. 1998). Such pressures might extend to analysts at broker IPOs through either explicit incentives (e.g., bonuses or equity-based compensation tied to firm performance that benefits from biased research) or implicit incentives (e.g., changes in corporate culture).³ In addition, analysts largely manage their own “franchise,” so individual reputation concerns may provide a disciplining mechanism that consistently encourages high-quality research regardless of any influence from the IPO (Fang and Yasuda 2009). Analysts’ reputations are closely associated with their reports, and investors can hold analysts accountable for low-quality research. Thus, the impact of the IPO process on analysts’ forecast quality is ultimately an empirical question.

We examine whether analyst earnings forecast quality changes during the period surrounding a broker IPO, which we identify as the two-year period centered on the IPO date.⁴ We focus on earnings forecasts because they are revised frequently, have a clear benchmark for evaluating performance (reported earnings), and provide a continuous measure of quality (i.e., bias or accuracy) with substantial variation. We measure forecast quality as *ex post* optimistic bias and accuracy. Optimistic bias is a useful metric for assessing analyst quality, with higher levels representing lower-quality forecasts, potentially attributable to investment banking and trading incentives (e.g., Cowen et al. 2006). Accuracy (i.e., absolute forecast error) is another well-established metric of forecast quality, as prior studies find that although accurate forecasts are beneficial to capital market participants (Clement and Tse 2003), they require greater resources (Clement 1999) and may lead to sacrifices in other business interests (Pacelli 2019).⁵

To test our prediction, we use data from Thomson One to identify IPOs in the financial services industry. The intersection of the ThomsonOne IPO data with I/B/E/S analyst data provides a sample of 23 broker IPOs that meet our sample criteria from 1985–2007.⁶ Our primary empirical analyses examine changes in individual analysts’ forecast bias or accuracy during the IPO process. In these tests, we use a generalized difference-in-difference (DiD) methodology to examine changes in one-year-ahead earnings forecast bias or accuracy in the two-year period centered on the IPO event (i.e., years $t - 1$ and $t + 1$). We focus on analysts employed at IPO brokers during this time (hereafter “IPO analysts”) and compare changes in forecast properties to those of all other analysts (“control analysts”). We include analyst and covered firm-year fixed effects in the models to reduce the possibility that our tests are confounded by alternative explanations related to the

covered firms (e.g., need for financing), analyst-specific characteristics (e.g., talent or inherent optimism), or time-varying economic conditions. We estimate our models using the full panel of forecasts archived by I/B/E/S from 1982 (the first year when analyst data are available from I/B/E/S) to 2012, which is five years after our last broker IPO. Our main sample includes 796,580 firm-year observations with analyst coverage.

Our primary results indicate that IPO analysts produce higher-quality forecasts around their broker IPOs. We document a decrease in forecast bias for IPO analysts that is roughly 6.4 basis points of price compared with that of control analysts. Similarly, we find a reduction in absolute forecast error of approximately 5.4 basis points of price for IPO analysts relative to control analysts. These initial results confirm our predicted increase in forecast quality and suggest that the IPO process has a significant positive impact on the equity research activities of the brokerage.

We next conduct a broad set of additional analyses to assess the robustness of our main findings. In the first set of analyses, we examine the timing dynamics of the IPO effect by examining pretrends and whether the effect is transitory. We find that the differences in forecast quality between broker IPO and control analysts are insignificant in the two-year period immediately preceding the IPO. Further, when we control for pretrend effects, we find our main results continue to hold, reinforcing our inference about a marked decrease in analyst optimistic bias and forecast errors for IPO analysts during the IPO process. However, forecast quality only appears to increase in the short window surrounding the IPO event, suggesting that IPOs have a transitory effect on analysts’ research quality.

In the second set of tests, we assess the robustness of our findings to alternative specifications and fixed effect structures that control for different types of unobserved heterogeneity. In these tests, we examine models with broker fixed effects to account for time-invariant brokerage policies (e.g., culture), analyst-broker fixed effects to account for individual analysts’ employment periods, and analyst-covered firm fixed effects to control for the analyst’s relationship with the covered firm.⁷ Across all specifications, the results consistently indicate that IPOs are associated with significant reductions in forecast bias and error, with similar economic magnitudes to those of our primary model.⁸

Finally, we assess the robustness of our forecast bias result to an alternative research output, stock recommendation revisions. Consistent with our results based on earnings forecasts, we find that IPO analyst recommendation revisions also become significantly less optimistic during the IPO period.

Overall, the evidence from our primary and supplemental analyses consistently supports our prediction that the IPO process improves analysts’ forecast quality. This is the key contribution of our paper. Identifying the

specific factors that drive this effect proves more challenging because the internal effects of the IPO on employees are largely unobservable to researchers. For example, we are unable to measure specific resources allocated to research departments during IPO events or equity compensation granted to analysts as part of the IPO event. This challenge notwithstanding, our remaining analyses attempt to provide additional evidence on potential factors that may impact the changes in analyst forecast quality that we observe during the IPO process. To be clear, we view these analyses as exploratory and highlight preliminary evidence of potential factors at play.

We start by examining whether brokerage houses invest more in research resources around the IPO. Ideally, we could measure such investments by examining internal budgets assigned to research departments. Given that such data are not available, we measure research investments using several coarse proxies that may be positively correlated with equity research investment and are also similar in spirit to those used in prior studies examining analyst resources (see, e.g., Clement 1999). These proxies include the number of firms covered, the number of analysts employed, and the number of forecasts produced by the brokerage in a given year. Although we find some evidence of an increase in equity research resources during the IPO process at the univariate level, we find that this effect dissipates in the presence of the IPO treatment main effect or with the inclusion of brokerage fixed effects.

Second, we explore the possibility that IPOs change the composition of employees by encouraging higher-quality employees to stay throughout the IPO process. To do this, we construct an indicator variable that captures whether an analyst stays with the broker throughout the entire two-year IPO window and regress this measure on our forecast quality measures (bias and accuracy) as well as an indicator for whether the analyst is an All-star analyst. The results provide strong evidence that All-star analysts are more likely to stay through the IPO process and some evidence that analysts with higher forecast quality also stay through the IPO process. We then examine whether IPO-related changes in quality are influenced by analysts who stay through the entire IPO. We re-estimate our baseline results with two separate treatments, one for analysts who stay through the IPO and another for analysts who leave during the IPO. We find that our results are concentrated among the analysts who remain at the brokerage throughout the entire IPO period. Together, these results suggest that higher-quality analysts stay through the IPO and that such analysts are the primary contributors to the observed improvements in forecast quality.

Finally, we explore the possibility that an IPO introduces enhanced scrutiny that disciplines analysts to produce higher-quality forecasts. We begin by partitioning our sample into two scrutiny regimes—the pre-Global

Settlement regime (1982–2002) and the post-Global Settlement regime (2003–2012). The rationale behind this analysis is that the increase in scrutiny of equity research departments after Global Settlement increased the overall objectivity of analyst reports (e.g., Kadan et al. 2009), thus potentially reducing any transitory effects of increased scrutiny around IPOs. Consistent with this argument, we find that our bias results are concentrated within the pre-Global Settlement period. We then examine whether our main results are moderated by analyst or brokerage characteristics that are likely to enhance the scrutiny effect. With respect to analyst characteristics, we find some evidence that our results are stronger among All-star analysts and analysts with high brokerage-specific experience, as such analysts are generally more visible and thus likely to face greater scrutiny. With respect to brokerage characteristics, we collect data on brokerages' business models similar to Cowen et al. (2006) and find that our effects are concentrated among investment banks as opposed to pure research firms. This result is consistent with investment banks facing increased scrutiny as they are more visible and face well-documented conflicts that can lead to biased research (e.g., Lin and McNichols 1998). Overall, the evidence from these analyses provide some support for the scrutiny and retention channels but less supportive evidence for the resources channel. We again exercise caution in overextrapolating these findings, however, because our tests rely on indirect proxies for internal effects of IPOs that we cannot directly observe.

The primary contribution of this study is to the literature on the determinants of analysts' research quality. In particular, a long line of literature finds that research quality is lower in the presence of covered firms' investment banking activities (e.g., Lin and McNichols 1998, Michaely and Womack 1999, Dechow et al. 2000, Irvine 2004, Jackson 2005, Bradshaw et al. 2006, Jacob et al. 2008). Our study extends this literature by demonstrating that the brokerage house's own IPO has implications for analysts' forecast quality, which temporarily increases as evidenced by declines in bias and error. Relatedly, our study also contributes to the literature examining determinants of analysts' forecast accuracy specifically (e.g., Clement 1999, Jacob et al. 1999, Brown et al. 2019, Pacelli 2019).

More broadly, our study offers a unique perspective on how IPOs can influence employee behavior. The analyst setting allows us to obtain employee-specific data, both before and after the IPO, that are typically unobservable to researchers in other IPO settings, including detailed information on the analysts and their output across time and employers. Prior studies exclusively focus on *firm*-level behavior (Teoh et al. 1998, Heaton 2002, Kedia et al. 2014, Ertimur et al. 2018), and there is limited empirical evidence on more granular effects of going public throughout an organization. Our findings provide some insight into these effects and

suggest that individual analysts provide higher-quality forecasts around their own brokers' IPOs.

2. Hypothesis Development

A primary function of investment banks is to help clients raise capital. A large literature in accounting and finance has documented the potential conflicts of interest that capital-raising activities have on analysts' research.⁹ Early research in this field provides compelling evidence that analysts face conflicts of interest. For example, Dugar and Nathan (1995) find that investment banks issue more optimistic forecasts and stock recommendations. Lin and McNichols (1998) document a higher degree of optimism for long-term growth forecasts and recommendations of analysts whose broker provides investment banking services for the covered firm. Similarly, Michaely and Womack (1999) find that unaffiliated analysts' buy recommendations outperform those of affiliated analysts. Ljungqvist et al. (2006) find that optimistic research benefits brokers by helping them attract co-underwriter management appointments and more lucrative deals. And finally, O'Brien et al. (2005) provide evidence that affiliated analysts are sluggish in downgrading recommendations from a buy rating.

In recent years, regulators have taken actions to reduce conflicts of interest. Notably, the Global Settlement was an enforcement agreement reached in 2003 between the Securities and Exchange Commission, Financial Industry Regulatory Authority, and New York Stock Exchange that involved 10 of the largest investment banks receiving nearly \$1.5 billion of fines related to conflicts of interest and biased research. In addition to these sanctions, the Global Settlement also introduced new regulations aimed at curbing bias, such as requiring analysts to disclose the outstanding levels of buys, holds, and sell recommendations in their portfolio.¹⁰

Although prior research focuses extensively on the capital raising activities of investment banks' clients, we are not aware of any study that examines the effects of a banks' own capital raising activities on its analysts' research. Our objective is to fill this gap in the literature by testing our central prediction that forecast quality will increase during broker IPOs. This prediction is motivated by several factors. First, the IPO event may encourage the brokerage house to invest more in equity research, which often represents a cost center for brokerage firms (Merkley et al. 2017). The additional capital can be used to improve the resources available to analysts to conduct their research (e.g., tools, technologies, support personnel, etc.) and ultimately result in increased forecast quality. This argument is consistent with Clement (1999), who finds a positive relationship between analyst forecast accuracy and resources. Thus, it is possible that the brokerage house may anticipate a successful IPO and ramp up its investment in equity research in the period immediately preceding the IPO.

Second, an IPO may provide a currency (i.e., equity shares) for attracting and retaining the best analysts. For example, prior research finds that stock compensation can help improve employee retention (Kole 1997, Oyer and Schaefer 2005, Ellingsen and Kristiansen 2022). Thus, the IPO process may also shift the composition of analysts working at the brokerage house. In particular, high-performing employees with attractive outside options may delay their departures from the firm in hopes of personally profiting from the IPO (e.g., equity interests).

Third, the IPO process introduces additional scrutiny and enhanced monitoring on the brokerage from external stakeholders, including investors, equity analysts, rating agencies, business press reporters, auditors, and lawyers. These parties reinforce regulatory initiatives for increased transparency (e.g., Sweeney 1994, Becker et al. 1998, Chung et al. 2002, Dyck et al. 2008, Yu 2008) and expose public companies to greater litigation risk (Lowry and Shu 2002, Badertscher et al. 2014). In the context of financial reporting, Ball and Shivakumar (2008) provide evidence that IPO firms report more conservatively because they are held to a higher reporting standard. The authors attribute this finding to enhanced scrutiny. Cecchini et al. (2012) provide similar evidence that IPO firms are more conservative with estimates reflected in the allowance for uncollectible accounts. Thus, increased forecast quality around an IPO can potentially be attributed to enhanced scrutiny and monitoring.

We note, however, that an increase in forecast quality is also not obvious *ex ante*. IPOs generate strong economic incentives for firms to report profitability (Teoh et al. 1998). Such pressures may naturally extend to nonexecutive employees, such as analysts, within a firm. Because research departments have traditionally been funded indirectly through institutional business (e.g., investment banking and trading fees), analysts are alleged to optimistically bias forecasts to support these business lines (Gu et al. 2013). Thus, the incentive to issue biased research may be even more salient to analysts just around the IPO of their brokerage when pressures to hit short-term performance targets become even more important. The incentives and pressures to focus on building these business lines could either be explicitly communicated through compensation structures or implicitly through a corporate culture that focuses on short-run results (e.g., Guiso et al. 2015, Pacelli 2019).

We also acknowledge that it is possible that broker IPOs may have no influence on analyst forecast quality. Prior studies find that personal reputation concerns are highly effective in disciplining sell-side analysts (Jackson 2005, Fang and Yasuda 2009). Because analysts effectively sign their reports, changes in objectivity around IPOs could potentially threaten the analyst's credibility with clients and harm future career outcomes. Thus, analysts' desire to maintain their own reputations may

constrain the extent to which their brokerage's IPO can influence their actions. Overall, the credible null is that analysts' forecast quality remains unchanged around broker IPOs.

3. Sample and Research Design

3.1. Sample Construction

We use several data sets to construct our sample. We begin by identifying IPOs in the financial services industry by searching the ThomsonOne database for all U.S. IPOs in Standard Industrial Classification industry groups 6211 (security brokers, dealers, and flotation) and 6282 (investment advice) during the 1985–2014 period.¹¹ Our focus on these particular industry groups is motivated by prior studies that use industry group 6211 when examining broker mergers and closures (Hong and Kacperczyk 2010), whereas industry group 6282 captures dedicated equity research firms. We manually match the issuer name in ThomsonOne to I/B/E/S broker names using the I/B/E/S broker translation file.

In Table 1, we provide a list of our final sample of 23 broker IPOs along with the IPO date. For each IPO, we verify the dates and details by manually collecting S-1 filings from the SEC EDGAR website and relevant news articles from Factiva. The IPOs in our sample are well distributed over time, with eight occurring in the 1980s, nine in the 1990s, and six in the 2000s.

We follow prior literature and obtain from the I/B/E/S detail file the last one-year-ahead annual earnings forecast issued by each analyst during the 11-month period ending 30 days before each covered firm's fiscal

year end date (Clement 1999). We include forecasts from 1982 (the start of the I/B/E/S detail file) through 2012 (five years after our last IPO event). We then retain all forecasts with necessary data to calculate the forecast variables (i.e., bias and accuracy) and control variables from I/B/E/S, as described below. We exclude forecasts from analysts at IPO brokers that are issued within one month of the IPO date (i.e., either before or after the IPO date).¹² We also remove observations where the covered firm has a missing market value of equity, a negative book-to-market ratio, or a stock price less than \$3 (Gu and Wu 2003). Finally, we require forecasts from at least two analysts for a given firm-year. Our final sample contains 796,580 analyst-firm-year observations. Within this sample, the treatment sample of forecasts issued during the year prior to and following the IPO contains 11,362 forecasts issued by 650 unique analysts.

3.2. Empirical Models

We test our primary prediction that analysts' forecast quality increases during the broker's IPO process using a generalized difference in differences (DiD) model (see, e.g., Christensen et al. 2016, Bourveau et al. 2018). This model uses the full panel of analyst forecasts available in I/B/E/S during the 1982–2012 period. We specify the model as follows:

$$\begin{aligned} ForQual_{ijbt} = & \alpha_1 IPOTreat[-1, +1]_{bt} + \alpha_2 BrokerSize_{bt} \\ & + \alpha_3 Horizon_{ijt} + \alpha_4 Coverage_{it} \\ & + \alpha_5 Experience_{ijt} + \beta FixedEffects_{ij \times t} + \varepsilon_{ibjt}. \end{aligned} \quad (1)$$

In Equation (1), i indexes analysts, b indexes broker (employing analyst i), j indexes the covered firm, and t indexes year. $ForQual$ is one of two alternative proxies of analyst forecast quality: *Bias* and *Error*. *Bias* is the analyst's last forecast for the year minus the covered firm's actual earnings, scaled by stock price measured two days prior to the forecast issuance date, and multiplied by 100. *Error* is the absolute value of the analyst's last forecast for the year minus the covered firm's actual earnings, scaled by stock price measured two days prior to the forecast issuance date, and multiplied by 100. Lower values of *Bias* or *Error* indicate higher-quality forecasts. The variable of interest in Equation (1) is the generalized DiD estimator $IPOTreat[-1, +1]$, which is an indicator variable that takes the value of one for forecasts issued by analysts at IPO brokerages during the two-year period centered on the IPO date and zero otherwise. This variable can be viewed as an interaction term for analysts employed by IPO brokerages and a time period indicator. The main effects for these variables are subsumed by our fixed effect structure, as discussed in the next paragraph. A negative and significant coefficient on $IPOTreat[-1, +1]$ provides

Table 1. IPO Events

Broker name	IPO date
Bear Stearns Cos Inc	10/29/1985
First Albany Cos Inc	12/11/1985
Alex Brown Inc	02/28/1986
Morgan Stanley Group Inc	03/21/1986
Rodman & Renshaw Capital Group	05/29/1986
Ryan Beck & Co	06/27/1986
Oppenheimer Capital LP	07/01/1987
The Charles Schwab Corp	09/23/1987
Southwest Securities Group Inc	10/11/1991
Dean Witter Discover & Co	02/22/1993
Donaldson Lufkin & Jenrette	10/25/1995
Hambrecht & Quist Group Inc	08/09/1996
Conning Corp	12/15/1997
Ragen MacKenzie Group Inc	06/22/1998
Goldman Sachs Group Inc	05/03/1999
Wit Capital Group Inc	06/04/1999
TD Waterhouse Group Inc	06/23/1999
Instinet Group Inc	05/17/2001
Lazard Ltd	05/04/2005
Thomas Weisel Partners Group	02/01/2006
KBW Inc	11/08/2006
JMP Group Inc	05/10/2007
FBR Capital Markets Corp	06/07/2007

Note. This table details our final sample of broker IPOs.

evidence consistent with our prediction that the IPO process coincides with higher-quality forecasts.

We include in Equation (1) an extensive fixed effect structure to help isolate the effect of going public on forecast quality and reduce the plausibility of alternative explanations associated with characteristics of the analyst, covered firm, or time. Our primary specification controls for time-invariant characteristics of the analyst through analyst fixed effects as well as time-varying characteristics of the covered firm through covered firm $\times$ year fixed effects. The covered firm $\times$ year fixed effects are particularly powerful as they control for unobservable characteristics of the firm that vary over time that might influence bias, such as the need for financing. We consider alternative fixed effect structures in robustness tests discussed below.

We also include in Equation (1) a set of control variables that relate to various characteristics of the analyst and the broker to capture any remaining residual variance related to temporal changes in such factors. More specifically, we control for broker size (*BrokerSize*), defined as the natural

log of the total number of analysts employed by the broker of interest.¹³ We further control for analyst forecast horizon (*Horizon*), defined as the natural log of the number of days between the forecast issuance date and the fiscal period end date. We also control for two important analyst characteristics. *Coverage* is the natural log of the total number of firms covered by the analyst. *Experience* is the natural log of the number of years of experience the analyst has been covering the firm. We winsorize all continuous variables at the top and bottom 1% to reduce the influence of outliers, and all variables are formally defined in the appendix. Finally, we cluster standard errors by covered firm-year.

3.3. Descriptive Statistics

In panel A of Table 2, we report descriptive statistics for the variables used in our models. We report unlogged values of the variables for ease of interpretation. On average, we find that annual earnings forecasts are optimistically biased at approximately 0.855% of stock price.

Table 2. Descriptive Statistics

Panel A: Descriptive statistics						
	N	Mean	Std. dev	25th	Median	75th
<i>PreTreat</i> [−3, −2]	796,580	0.013	0.115	0.000	0.000	0.000
<i>IPO</i> <i>Treat</i> [−1, +1]	796,580	0.014	0.119	0.000	0.000	0.000
<i>PostTreat</i> [+2, +3]	796,580	0.016	0.125	0.000	0.000	0.000
<i>PostTreat</i> [4+]	796,580	0.088	0.284	0.000	0.000	0.000
<i>Bias</i>	796,580	0.855	4.914	−0.173	0.000	0.312
<i>Error</i>	796,580	1.796	6.286	0.059	0.223	0.834
<i>BrokerSize</i>	796,580	60.747	58.433	19.000	43.000	81.000
<i>Horizon</i>	796,580	114.065	81.333	58.000	73.000	155.000
<i>Coverage</i>	796,580	18.321	13.737	10.000	16.000	22.000
<i>Experience</i>	796,580	2.102	3.289	0.000	1.000	3.000
<i>RecRevision</i>	289,164	−0.119	1.311	−1.000	0.000	1.000
<i>Num_Firms</i>	6,567	154.141	233.762	17.000	58.000	176.000
<i>Num_Analysts</i>	6,567	18.171	26.441	3.000	9.000	20.000
<i>Num_Forecasts</i>	6,567	556.450	945.154	48.000	169.000	598.000
<i>Stay</i>	469	0.770	0.421	1.000	1.000	1.000
<i>Avg_Bias</i>	469	0.726	1.663	−0.056	0.175	0.896
<i>Avg_Error</i>	469	1.557	2.193	0.278	0.653	1.605
<i>All-star</i>	469	0.288	0.453	0.000	0.000	1.000
Panel B: IPO and non-IPO means						
	<i>PreTreat</i> [−3, −2]	<i>IPO</i> <i>Treat</i> [−1, +1]	<i>PostTreat</i> [+2, +3]	<i>PostTreat</i> [4+]	Control	
Max N (forecast sample)	10,707	11,362	12,603	70,295	664,936	
<i>Bias</i>	0.751	0.985	1.164	0.696	0.855	
<i>Error</i>	1.559	1.862	2.271	1.727	1.789	
<i>BrokerSize</i>	43.662	51.556	60.613	81.838	59.483	
<i>Horizon</i>	115.276	112.811	112.070	109.343	114.348	
<i>Coverage</i>	19.055	19.867	20.125	16.683	18.320	
<i>Experience</i>	1.710	2.012	2.286	2.413	2.102	
<i>RecRevision</i>	−0.063	−0.188	−0.063	−0.122	−0.119	
<i>Num_Firms</i>	353.270	376.488	386.225	386.225	138.270	
<i>Num_Analysts</i>	31.162	34.244	37.600	37.600	16.635	
<i>Num_Forecasts</i>	1,236.700	1,336.410	1,462.130	1,462.130	496.054	

Notes. This table provides descriptive statistics for the variables used in our analyses. Unlogged values of the variables are reported for ease of interpretation. Variable definitions are provided in the appendix. Std. dev., standard deviation.

Analysts in our sample issue forecasts approximately 114 days before the fiscal year end. Further, analysts work for brokers that employ, on average, 61 analysts, cover approximately 18 firms, and have a little over two years of experience covering the firm of interest.¹⁴

In Table 2, panel B, we present variable means across IPO and non-IPO observations for various windows. For this panel, non-IPO (control) observations consist of brokers that never go public during our sample period. Consistent with IPOs generating an increase in resources for the equity research division, we find that from the pre-IPO window to the IPO window, the number of analysts increases from 31 to 34, the total number of covered firms at the brokerage increases from 353 firms to 376, and the number of forecasts increases from 1,237 to 1,462. We also note, however, that these increases may simply reflect a general growth in the equity research division, as we continue to see increases in these metrics well beyond the IPO period in the *IPO[4+]* window. This highlights the importance of drawing inferences from the results of our generalized DiD models, which explicitly control for time trends. With respect to forecast bias and accuracy, we find that both metrics increase from the pre-IPO window to the IPO window; but we again refrain from drawing inferences based on these univariate statistics. With respect to the control sample, they appear to have comparable levels of bias and accuracy, with means of 0.855 and 1.789, respectively.

4. Does Going Public Increase Analyst Forecast Quality?

4.1. Primary Hypothesis Tests

In Table 3, we provide the estimation results from Equation (1), which is the model we use to test our

Table 3. Broker IPOs and Analyst Forecast Quality

Dependent variable	(1) <i>Bias</i>	(2) <i>Error</i>
<i>IPOTreat</i> [−1, +1]	−0.064*** (−2.72)	−0.054** (−2.35)
<i>BrokerSize</i>	−0.009 (−1.64)	−0.016*** (−2.87)
<i>Horizon</i>	0.378*** (37.57)	0.547*** (54.78)
<i>Coverage</i>	0.015*** (2.91)	0.011* (1.93)
<i>Experience</i>	−0.016*** (−4.19)	−0.013*** (−3.31)
Analyst Fixed Effects	Yes	Yes
Firm-Year Fixed Effects	Yes	Yes
<i>N</i>	794,932	794,932
Adjusted <i>R</i> ²	0.797	0.875

Notes. This table presents results from estimating the generalized difference-in-differences Equation (1). T-statistics are reported in parentheses, and standard errors are clustered by firm-year. All *p*-values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

primary hypothesis. In column (1), we use *Bias* as the dependent variable; in column (2), we use *Error*. We find that both models exhibit strong explanatory power with R-squared statistics of 0.797 and 0.875 in columns (1) and (2), respectively. In both columns, we find that the coefficient on *IPOTreat*[−1, +1] is negative and significant at the 5% or 1% level. These results suggest that analyst forecasts become less optimistically biased and more accurate during the IPO event window. The magnitudes on the coefficients also appear to be economically meaningful. The coefficient on *IPOTreat*[−1, +1] is −0.064 in column (1), which indicates that broker IPOs are associated with a reduction in analyst optimism of 6.4 basis points of price. Similarly, the increase in forecast accuracy presented in column (2) is slightly lower at 5.4 basis points of price.

We next estimate several alternative specifications of Equation (1) to examine the robustness of our primary results. First, we estimate Equation (1) using several alternative specifications and fixed effect structures. First, we add two additional control variables to the model related to broker characteristics. These *Additional Controls* are the natural log of the number of firms that the brokerage covers during the year (*Num_Firms*) and the natural log of the number of one-year-ahead earnings forecasts that the brokerage issues during the year (*Num_Forecasts*). Second, we add broker fixed effects to the model to control for unobserved time-invariant heterogeneity at the broker level.¹⁵ Third, we consider analyst-broker interactive fixed effects to hold constant any particular analyst-brokerage pairing. For example, this specification separately exploits variation in bias from analyst A's forecasts while employed by Goldman Sachs from those while analyst A is employed by Morgan Stanley. Fourth, we consider analyst-covered firm interactive fixed effects to hold constant the effects within a particular analyst-covered firm relationship (e.g., analysts A's forecasts of Microsoft). The analyst-covered firm fixed effects thus control for the analyst's relationship with the covered firm over time. In our final set of robustness tests, we estimate Equation (1) using several alternative two-way standard error clustering methods, including clustering by (i) covered-firm and year, (ii) broker and year, and (iii) analyst and year.

In Table 4, we present the results of the robustness tests just described for our primary Equation (1). In panel A of Table 4, we present the specifications using additional controls and alternative fixed effect structures; in panel B, we present the results using alternative standard error clustering. We find that our primary results are very robust to these alternative specifications. The coefficient on *IPOTreat*[−1, +1] is negative in all 14 specifications across the two panels and is statistically significant in 13 of the 14 specifications.

In our second robustness test, we assess for potential pretrends and control for any possible changes in analyst behavior prior to the IPO period. We add an indicator

Table 4. Broker IPOs and Analyst Forecast Quality (Alternative Fixed Effects, Controls, and Clustering)

Panel A: Controls and fixed effects								
Dependent variable	(1) Bias	(2) Error	(3) Bias	(4) Error	(5) Bias	(6) Error	(7) Bias	(8) Error
<i>IPOTreat</i> [−1, +1]	−0.065*** (−2.76)	−0.055** (−2.39)	−0.063*** (−2.65)	−0.053** (−2.22)	−0.061** (−2.40)	−0.062** (−2.46)	−0.037 (−1.44)	−0.042* (−1.77)
Primary controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	No	No	No	No	No	No
Analyst Fixed Effects	Yes	Yes	Yes	Yes	No	No	No	No
Firm-year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Broker Fixed Effects	No	No	Yes	Yes	No	No	Yes	Yes
Analyst-broker Fixed Effects	No	No	No	No	Yes	Yes	No	No
Analyst-firm Fixed Effects	No	No	No	No	No	No	Yes	Yes
<i>N</i>	794,932	794,932	794,880	794,880	792,253	792,253	691,575	691,575
Adjusted <i>R</i> ²	0.797	0.875	0.798	0.875	0.799	0.876	0.812	0.884
Panel B: Clustering								
Dependent variable	(1) Bias Firm and year	(2) Error Firm and year	(3) Bias Broker and year	(4) Error Broker and year	(5) Bias Analyst and year	(6) Error Analyst and year		
<i>IPOTreat</i> [−1, +1]	−0.064*** (−2.77)	−0.054** (−2.31)	−0.064*** (−3.21)	−0.054** (−2.09)	−0.064*** (−2.82)	−0.054** (−2.29)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes		
Analyst Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes		
Firm-year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes		
<i>N</i>	794,932	794,932	794,932	794,932	794,932	794,932		
Adjusted <i>R</i> ²	0.797	0.875	0.797	0.875	0.797	0.875		

Notes. This table presents results from estimating the generalized difference-in-differences Equation (1) using alternative fixed effects, controls, and alternative standard error clustering. T-statistics are reported in parentheses. All *p*-values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

variable *IPOTreat*[−3, −2] to Equation (1) that takes the value of one for forecasts issued by analysts at IPO brokerages during the two-year period ending one year before the IPO date and zero otherwise. We specify the model as follows:

$$\begin{aligned} \text{ForQual}_{ijt} = & \alpha_1 \text{IPOTreat}[-3, -2]_{bt} \\ & + \alpha_2 \text{IPOTreat}[-1, +1]_{bt} + \alpha_3 \text{BrokerSize}_{bt} \\ & + \alpha_4 \text{Horizon}_{ijt} + \alpha_5 \text{Coverage}_{it} + \alpha_6 \text{Experience}_{ijt} \\ & + \beta \text{FixedEffects}_{ijt} + \varepsilon_{ijt}, \end{aligned} \quad (2)$$

where all variables are as defined previously.

Next, we expand Equation (2) to assess posttrends by including two additional indicator variables. *IPOTreat*[+2, +3] takes the value of one for forecasts issued by analysts at IPO brokerages during the two-year period starting one year after the IPO date, and zero otherwise. *IPOTreat*[4+] takes the value of one for all forecasts issued by analysts at IPO brokerages starting four years after the IPO date, and zero otherwise. This specification allows us to assess how long-lived the effects of the IPO are. We specify the model as follows:

$$\begin{aligned} \text{ForQual}_{ijt} = & \alpha_1 \text{IPOTreat}[-3, -2]_{bt} \\ & + \alpha_2 \text{IPOTreat}[-1, +1]_{bt} + \alpha_3 \text{IPOTreat}[+2, +3]_{bt} \\ & + \alpha_4 \text{IPOTreat}[4+]_{bt} + \alpha_5 \text{BrokerSize}_{bt} \\ & + \alpha_6 \text{Horizon}_{ijt} + \alpha_7 \text{Coverage}_{it} + \alpha_8 \text{Experience}_{ijt} \\ & + \beta \text{FixedEffects}_{ijt} + \varepsilon_{ijt}, \end{aligned} \quad (3)$$

where all variables are defined previously.

In Table 5, we provide the estimation results for Equations (2) and (3) that introduce controls for pre-IPO and for pre- and post-IPO effects, respectively. We present the Equation (2) results for *Bias* and *Error* in columns (1) and (2), respectively; we present the Equation (3) results in columns (3) and (4). We first note that the coefficient on *IPO*[−1, 1] is negative and significant at the 5% or 1% level across all four models. Further, the magnitudes of the coefficients remain similar to those reported in our primary analysis in Table 3, ranging from 5.7–7.5 basis points of price. We also note that the coefficient on the pretrend indicator variable, *IPO*[−3, −2], is insignificantly different from zero in all specifications. This provides evidence that forecasts from analysts at IPO and non-IPO brokers are not statistically different in the years immediately prior to the IPO event; thus, we fail to find evidence of significant pretrend effects. Finally, in columns (3) and (4), we note that the coefficients on the posttrend indicator variables, *IPOTreat*[+2, +3] and *IPOTreat*[4+], are also all statistically insignificant, with one exception (a marginally positive coefficient on *IPOTreat*[4+] in column (4)). We plot the coefficients on each of the Equation (3) indicator variables from columns (3) and (4), along with their 90% confidence intervals, in Figure 1 (*Bias*) and Figure 2 (*Error*). We observe a marked change in analyst forecast bias in the two years around the IPO, but not before or after, and similar, although slightly weaker evidence, with respect to forecast error. This provides additional support for our inference that the observed improvements in forecast

Table 5. Broker IPOs and Analyst Forecast Quality (Pre- and Posteffects)

Dependent variable	(1) Bias	(2) Error	(3) Bias	(4) Error
<i>PreTreat</i> [−3, −2]	−0.024 (−0.98)	−0.012 (−0.51)	−0.027 (−1.11)	−0.014 (−0.56)
<i>IPO</i> <i>Treat</i> [−1, +1]	−0.069*** (−2.81)	−0.057** (−2.35)	−0.075*** (−2.92)	−0.061** (−2.45)
<i>PostTreat</i> [+2, +3]			−0.028 (−0.96)	−0.043 (−1.51)
<i>PostTreat</i> [4+]			0.006 (0.36)	0.031* (1.93)
<i>BrokerSize</i>	−0.009* (−1.65)	−0.016*** (−2.88)	−0.009* (−1.67)	−0.017*** (−2.97)
<i>Horizon</i>	0.378*** (37.57)	0.547*** (54.78)	0.378*** (37.56)	0.547*** (54.78)
<i>Coverage</i>	0.015*** (2.93)	0.011* (1.94)	0.016*** (2.94)	0.011* (1.94)
<i>Experience</i>	−0.016*** (−4.19)	−0.013*** (−3.31)	−0.016*** (−4.18)	−0.013*** (−3.32)
Analyst Fixed Effects	Yes	Yes	Yes	Yes
Firm-year Fixed Effects	Yes	Yes	Yes	Yes
<i>N</i>	794,932	794,932	794,932	794,932
Adjusted <i>R</i> ²	0.797	0.875	0.797	0.875

Notes. This table presents results from estimating the generalized difference-in-differences Equations (2) and (3). T-statistics are reported in parentheses, and standard errors are clustered by firm-year. All *p*-values are two tailed.

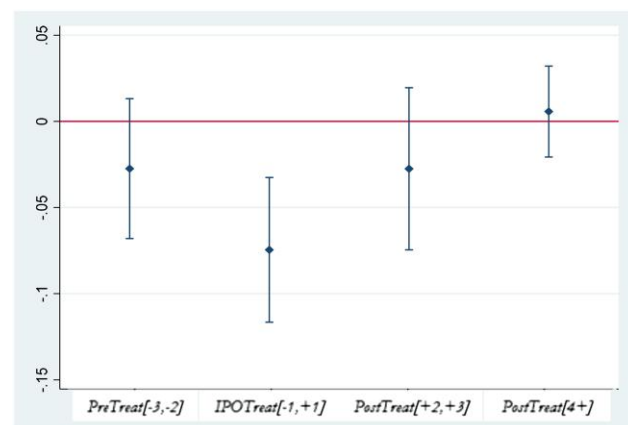
*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

quality are concentrated during the IPO event window. Overall, the results presented in Tables 3–5 provide consistent evidence that the IPO process coincides with an improvement in analyst forecast quality.

4.2. Tests Using Alternative Measures for Forecast Quality

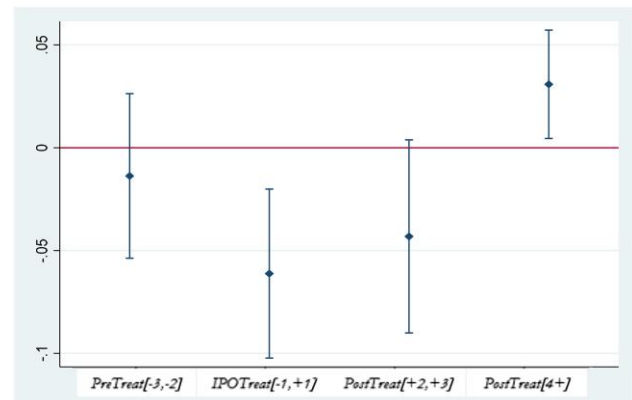
In our next test, we examine the robustness of our results to an important alternative analyst output, stock

Figure 1. (Color online) Trends in Analyst Forecast Bias Around IPOs



Note. This figure plots the treatment variable coefficients and related 90% confidence intervals from estimations of the generalized difference-in-differences Equation (3) for forecast bias.

Figure 2. (Color online) Trends in Analyst Forecast Error Around IPOs



Note. This figure plots the treatment variable coefficients and related 90% confidence intervals from estimations of the generalized difference-in-differences Equation (3) for forecast error.

recommendation revisions.¹⁶ Optimistic bias in analyst stock recommendations has been documented in numerous prior studies (e.g., Lin and McNichols 1998, Michaely and Womack 1999, Mehran and Stulz 2007) and indeed was one of the primary drivers for the Global Settlement.

We obtain analyst stock recommendations from I/B/E/S and define the variable *RecRevision* as the difference between an analyst's current recommendation and prior recommendation for the covered firm, where recommendations are coded on a five-point scale (1 = strong sell, 2 = sell, 3 = hold, 4 = buy, 5 = strong buy). Thus, positive values of *RecRevision* indicate upward revisions and negative values indicate downward revisions. We use a five-point rating scale because this was the rating scheme used by analysts through most of our IPO events, but note that our results are robust to using a three-point rating scale (Kadan et al. 2009). We then replace our dependent variable *Bias* with *RecRevision* in Equation (1). The model is again estimated with analyst and covered firm-year fixed effects. We note that our sample drops to 270,909 observations for this test because recommendations are far less frequently revised than are earnings forecasts.

We present the results from our recommendation revision tests in Table 6. Consistent with our primary results, we find a negative and significant (at the 1% level) coefficient on the IPO event indicator variable *IPO**Treat*[−1, +1], suggesting that the size of downward recommendation revisions is larger (or the size of upward revisions is smaller) in the IPO window. As a follow-up test, we examine the frequency of upward versus downward revisions. We find that the vast majority of recommendation revisions change the recommendation by one point (either up or down) on the five-point rating system, with downgrades constituting 43% of all revisions and upgrades constituting 35% of all revisions (untabulated). We further find that 2% of

Table 6. Broker IPOs and Stock Recommendations

Dependent variable	(1) <i>RecRevision</i>
<i>IPOtreat</i> [−1, +1]	−0.118*** (−4.74)
<i>BrokerSize</i>	0.004 (0.65)
<i>Coverage</i>	−0.017** (−2.39)
<i>Experience</i>	−0.004 (−1.11)
Analyst Fixed Effects	Yes
Firm-year Fixed Effects	Yes
<i>N</i>	270,909
Adjusted <i>R</i> ²	−0.033

Notes. This table presents results from estimating a modified version of the generalized difference-in-differences Equation (1) where the dependent variable is stock recommendation revisions (*RecRevision*). T-statistics are reported in parentheses, and standard errors are clustered by firm-year. All *p*-values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

all downward revisions drop the recommendation by more than two points on the rating scale, but only 1.5% of all upward revisions increase the recommendation by more than two points on the ratings scale. This evidence provides additional support for the idea that analyst optimism bias decreases around broker IPOs.

5. What Drives Changes in Analyst Forecast Quality During the IPO Process?

In this section, we explore factors that could potentially explain the increase in forecast quality around broker IPOs. We view these tests as exploratory and caution readers against drawing strong inferences because of the preliminary, and largely indirect, nature of the evidence presented. That is, internal effects of the IPO process on employees are unobservable to researchers. We consider three potential factors that are not mutually exclusive. These factors include changes in brokerage resources, changes in employee composition, and changes in brokerage scrutiny. We examine each in turn.

5.1. Investment in Equity Research

We first explore whether there is an influx of resources to the equity research division associated with the IPO that influences the observed increase in forecast quality. The new resources could be used to acquire new research tools, new data, or to increase the size of the analyst teams and support staff, all of which could potentially increase the quality of the research outputs produced by the division. To examine this possibility, we construct a new broker-year panel where the treated brokers are the 23 brokers listed in Table 1. We use this broker-level sample because the increase in resources

to the brokerage is at the broker level rather than at the individual analyst level as in our primary tests. This new sample consists of 6,567 broker-year observations. If the equity research division receives an increase in resources, then we would expect it to grow, which we measure using three broker characteristics, namely, the number of covered firms (*Num_Firms*), the number of analysts (*Num_Analysts*), and the number of forecasts issued (*Num_Forecasts*). Recall that we observe some evidence of an increase in resources using these three variables in Table 2, panel B. However, here we investigate this increase more rigorously using the following generalized DiD model at the broker-year level:

$$ResearchResources_{bt} = \alpha_1 IPOtreat[-1, +1]_{bt} + \beta FixedEffects_{bt} + \varepsilon_{bt}. \quad (4)$$

In Equation (4), the *b* subscript indexes brokerage house and *t* indexes calendar year. *ResearchResources* is either *Num_Firms*, *Num_Analysts*, or *Num_Forecasts* as defined above. The variable of interest is again *IPOtreat*[−1, +1]. We examine three specifications of Equation (4). In the first specification, we include year fixed effects only. In the second specification, we include year fixed effects as well as the IPO treatment brokerage main effect (*IPOtreat*) to account for fixed differences in resources between treatment and control brokerages. In the third specification, we include both year fixed effects and broker fixed effects. This specification is akin to a generalized DiD model and is similar in spirit to the specification used in our primary forecast analyses. A positive coefficient on α_1 would be consistent with a significant increase in equity research division resources around the time of the IPO.

We present the Equation (4) estimation results in Table 7. Although we do find some evidence of an increase in resources, this is only significant in columns (1)–(3), which omit either the IPO treatment main effect (*IPOtreat*) or broker fixed effects. Thus, we interpret this as very modest evidence of an increase in equity research division resources around the time of the IPO. In untabulated analyses, we examine this nonresult further by reviewing the use of the proceeds field in Securities Data Company for each of the broker IPOs in our sample. We find that no broker in our sample states that the proceeds of the offering will be used to support research activities. Together, this evidence suggests the observed increase in forecast quality around IPOs is unlikely to be explained by increases in equity research division resources.

5.2. Analyst Composition Tests

In our next test, we explore changes in the composition of employees at the brokerage and whether this influences the quality of the forecasts issued by the analysts at the brokerage. A brokerage house about to go public may be more likely to retain its best and most

Table 7. Broker IPOs and Brokerage Research Investment

Dependent variable	(1) Num_ Firms	(2) Num_ Analysts	(3) Num_ Forecasts	(4) Num_ Firms	(5) Num_ Analysts	(6) Num_ Forecasts	(7) Num_ Firms	(8) Num_ Analysts	(9) Num_ Forecasts
<i>IPOTreat</i> [−1, +1]	1.273*** (5.17)	0.921*** (4.67)	1.388*** (4.99)	−0.094 (−0.38)	−0.155 (−0.77)	−0.137 (−0.49)	0.069 (0.64)	−0.029 (−0.33)	0.072 (0.53)
<i>IPOTreat</i>				1.467*** (18.92)	1.154*** (18.59)	1.637*** (18.68)			
Broker Fixed Effects	No	No	No	No	No	No	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	6,567	6,567	6,567	6,567	6,567	6,567	6,374	6,374	6,374
Adjusted <i>R</i> ²	0.076	0.061	0.043	0.124	0.108	0.092	0.843	0.829	0.799

Notes. This table presents results from estimating the generalized difference-in-differences Equation (4) on a sample of broker-year level observations. T-statistics are reported in parentheses. All *p*-values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

productive employees through the IPO process because they can cash in on windfall compensation (e.g., equity compensation) resulting from going public. This may shift the research quality upwards during the IPO period because the right tail of the analyst distribution stays with the firm. This increase in the retention of the best analysts could potentially improve the average research quality at the brokerage, albeit only temporarily, as such employees may eventually depart.

To examine this possibility, we estimate the following linear probability model at the individual analyst level:

$$Stay_a = \alpha_1 AnalystForecastQuality_a + \alpha_2 BrokerSize_b + \alpha_3 Coverage_a + \alpha_4 Experience_a + \varepsilon_a. \quad (6)$$

In Equation (6), the *a* subscript indexes analyst and *b* indexes broker. The dependent variable *Stay* is an indicator variable set equal to one for analysts that remain at the brokerage during the entire two-year IPO window and to zero otherwise. *AnalystForecastQuality* is one of three independent variables. *Avg_Bias* is the average *Bias* of all forecasts issued by the analyst during the year prior to the IPO period (i.e., year *t* − 2). *Avg_Error* is the average *Error* measured over the same time period. *All-star* is an indicator set equal to one if the analyst is an All-star and to zero otherwise. All other variables are as defined previously. A significant negative coefficient on *Avg_Bias* or *Avg_Error* or a significant positive coefficient on *All-star* means that analysts that produce higher (lower) quality forecasts are more likely to remain (leave) at the firm as it enters into the IPO window.

We present the Equation (6) estimation results in Table 8. In column (1), we find a significant negative coefficient on *Avg_Bias*, indicating that higher-quality analysts are more likely to remain at the firm during the IPO window. In column (2), we also find that the coefficient on *Avg_Error* is negative but statistically insignificant. Finally, we find that the coefficient on *All-star* in column (3) is positive and significant. Thus, we find some evidence consistent with the idea that the retention of higher-quality analysts (i.e., those exhibiting

less bias and All-star distinctions) may be a contributing factor to the observed increase in analyst forecast quality around broker IPOs.

In an additional test, we explore whether IPO-related changes in quality are influenced by the analysts that stay through the entire IPO. We do this by reexamining our primary forecast analysis (i.e., Equation (1)) after partitioning the generalized DiD estimator *IPOTreat*[−1, +1] into two variables. *IPOTreat*[−1, +1]*_Stay* indicates treatment analysts that remain at the brokerage during the entire IPO window, and *IPOTreat*[−1, +1]*_Leave* indicates treatment analysts that leave the brokerage at some point during the IPO window. The model is as follows:

$$\begin{aligned} ForQual_{ijt} = & \alpha_1 IPOTreat[-1, +1]_{Stay_{bt}} \\ & + \alpha_1 IPOTreat[-1, +1]_{Leave_{bt}} \\ & + \alpha_2 BrokerSize_{bt} + \alpha_3 Horizon_{ijt} + \alpha_4 Coverage_{it} \\ & + \alpha_5 Experience_{ijt} \\ & + \beta FixedEffects_{ijt \times t} + \varepsilon_{ijt}. \end{aligned} \quad (7)$$

Table 8. Broker IPOs and Changes in Analyst Composition

Dependent variable = <i>Stay</i>	(1)	(2)	(3)
<i>Avg_Bias</i>	−0.024** (−2.07)		
<i>Avg_Error</i>		−0.008 (−0.84)	
<i>All-star</i>			0.109** (2.29)
<i>BrokerSize</i>	0.075*** (2.62)	0.078*** (2.70)	0.059* (1.96)
<i>Coverage</i>	0.007 (0.32)	0.005 (0.24)	−0.006 (−0.28)
<i>Experience</i>	0.007 (0.25)	0.010 (0.37)	0.001 (0.05)
<i>N</i>	469	469	469
Adjusted <i>R</i> ²	0.019	0.012	0.021

Notes. This table presents results from estimating Equation (6) on a sample of analyst-level observations. T-statistics are reported in parentheses. All *p*-values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

In Equation (1), i indexes analysts, b indexes broker (employing analyst i), j indexes the covered firm, and t indexes year. All variables are as defined previously.

We present the estimation results using these two indicator variables in Table 9 using *Bias* and *Error* as dependent variables in columns (1) and (2), respectively. We find that the forecast quality effects associated with broker IPOs are observed only in the set of analysts that stay at the brokerage during the entire IPO process. This provides corroborating evidence that the retention of higher-quality analysts may be contributing to the positive forecast quality effects we observe around IPOs.

5.3. Increased Scrutiny and Monitoring

In our final set of tests, we explore the possibility that increased scrutiny from various external stakeholders during the IPO processes is contributing to the observed forecast quality increases around broker IPOs. We first examine this question by exploiting an important scrutiny regime shift—the Global Settlement in 2002—that occurred during our sample period that increased the general level of scrutiny on brokerages (Kadan et al. 2009). This test leverages the idea that if scrutiny is an important factor at play, then any transitory impact of IPOs on analyst forecast quality will likely be reduced after Global Settlement. We test this conjecture by estimating Equation (1) separately during the pre-Global Settlement

regime (1982–2002) and the post-Global Settlement regime (2003–2012). If scrutiny is an important factor, then we expect the results to be stronger pre-Global Settlement.

We present the estimation results in Table 10. We find evidence of a significant IPO effect for *Bias* during the pre-Global Settlement regime only (column (1)). The coefficient on $IPO_{Treat[-1, +1]}$ is statistically insignificant in the other tests presented in columns (2)–(4). We also note that the coefficient is negative and larger in magnitude during the post-Global Settlement regime. Thus, overall, Table 10 provides only modest support for a scrutiny effect.

We next explore whether scrutiny is an important underlying factor by exploiting variation in analyst and brokerage characteristics. The idea is that certain analyst or brokerage characteristics are likely to enhance any scrutiny effects associated with broker IPOs, if such effects indeed exist. More specifically, we expect that All-star analysts and those with more experience at the brokerage are likely to face enhanced scrutiny. We also expect that analysts working for full service investment banks are more likely to face enhanced scrutiny compared with analysts working for research only brokerages. We explore these possibilities by examining the Equation (1) results after partitioning the generalized DiD estimator $IPO_{Treat[-1, +1]}$ into two variables. $IPO_{Treat[-1, +1]}_{HighScrutiny}$ indicates forecasts issued by treatment analysts that are All-stars, have above median experience at the brokerage, or work for a full service investment bank. $IPO_{Treat[-1, +1]}_{LowScrutiny}$ indicates forecasts issued by treatment analysts that are

Table 9. Broker IPOs, Analyst Forecast Quality, and Changes in Analyst Composition

Dependent variable	(1) <i>Bias</i>	(2) <i>Error</i>
$IPO_{Treat[-1, +1]}_{Stay}$	−0.073*** (−3.03)	−0.057** (−2.39)
$IPO_{Treat[-1, +1]}_{Leave}$	0.132 (1.32)	−0.001 (−0.01)
<i>BrokerSize</i>	−0.009 (−1.64)	−0.016*** (−2.87)
<i>Horizon</i>	0.378*** (37.56)	0.547*** (54.78)
<i>Coverage</i>	0.015*** (2.92)	0.011* (1.93)
<i>Experience</i>	−0.016*** (−4.19)	−0.013*** (−3.31)
Analyst Fixed Effects	Yes	Yes
Firm-year Fixed Effects	Yes	Yes
<i>N</i>	794,932	794,932
Adjusted R^2	0.797	0.875
Within Reg F-tests		
Diff	−0.205**	−0.056
F-Stat	(3.99)	(0.27)

Notes. This table presents results from estimating a modified version of the generalized difference-in-differences Equation (7), where treatment analysts are partitioned based on whether they stay with their brokerage throughout the IPO period. T-statistics are reported in parentheses, and standard errors are clustered by firm-year. All p -values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 10. Broker IPOs and Analyst Forecast Quality (Effects of Global Settlement)

Dependent variable	Pre-GS (1) <i>Bias</i>	(2) <i>Error</i>	Post-GS (3) <i>Bias</i>	(4) <i>Error</i>
$IPO_{Treat[-1, +1]}$	−0.060** (−2.22)	−0.038 (−1.56)	−0.080 (−1.54)	−0.084 (−1.50)
<i>BrokerSize</i>	−0.010 (−1.41)	−0.012* (−1.73)	0.004 (0.41)	−0.014 (−1.26)
<i>Horizon</i>	0.419*** (34.19)	0.482*** (41.80)	0.285*** (16.29)	0.657*** (34.24)
<i>Coverage</i>	0.014** (2.31)	0.019*** (3.08)	0.017 (1.34)	−0.007 (−0.55)
<i>Experience</i>	−0.014*** (−2.72)	−0.016*** (−2.94)	−0.019*** (−3.20)	−0.017*** (−2.62)
Analyst Fixed Effects	Yes	Yes	Yes	Yes
Firm-year Fixed Effects	Yes	Yes	Yes	Yes
<i>N</i>	485,497	485,497	308,847	308,847
Adjusted R^2	0.817	0.890	0.763	0.852

Notes. This table presents results from estimating the generalized difference-in-differences Equation (1), splitting the sample around Global Settlement. T-statistics are reported in parentheses, and standard errors are clustered by firm-year. All p -values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

not All-stars, have below median experience at the brokerage, or work for a pure research brokerage. The model is as follows:

$$\begin{aligned} ForQual_{ijt} = & \alpha_1 IPOTreat[-1, +1]_{HighScrutiny_{bt}} \\ & + \alpha_1 IPOTreat[-1, +1]_{LowScrutiny_{bt}} \\ & + \alpha_2 BrokerSize_{bt} + \alpha_3 Horizon_{ijt} + \alpha_4 Coverage_{it} \\ & + \alpha_5 Experience_{ijt} \\ & + \beta FixedEffects_{ijt} + \varepsilon_{ijt}. \end{aligned} \quad (8)$$

In Equation (8), i indexes analysts, b indexes broker (employing analyst i), j indexes the covered firm, and t indexes year. $IPOTreat[-1, +1]_{HighScrutiny}$ is one of three alternative variables. $IPOTreat[-1, +1]_{Star}$ is an indicator for All-star status; $IPOTreat[-1, +1]_{High_B_Exp}$ is an indicator for above median brokerage experience; and $IPOTreat[-1, +1]_{InvestmentBank}$ is an indicator for analysts working at full-service investment banks.¹⁷ Similarly, $IPOTreat[-1, +1]_{LowScrutiny}$ is one of three alternative variables. $IPOTreat[-1, +1]_{NonStar}$ is an indicator for non-All-star status; $IPOTreat[-1, +1]_{Low_B_Exp}$ is an indicator for below median brokerage experience; and $IPOTreat[-1, +1]_{PureResearch}$ is an indicator for analysts working at pure research brokerages. All other variables are as defined previously.

We present the Equation (8) estimation results in Table 11. Here we find some modest support for the scrutiny explanation. The coefficient on the DiD estimator is negative and statistically significant in five of the six regressions for analysts facing high scrutiny and in only one of the six regressions for analysts facing low scrutiny. The contrast in coefficients between the two groups is most apparent in columns (5) and (6), which exploit variation in scrutiny using the investment bank versus pure research firm distinction. However, none of these differences in coefficients are statistically significant.

6. Conclusion

We examine the effects of a broker going public on the forecast quality of the brokerage's sell-side equity analysts. Using a sample of 23 IPOs spanning three decades, we find that forecast quality improves among analysts at IPO brokers relative to analysts employed by other brokers during the IPO period. Specifically, we observe that during the IPO period, forecast bias decreases and forecast accuracy increases. These results are consistent across various robustness tests. We also provide some modest preliminary supporting evidence that IPOs introduce enhanced scrutiny that disciplines

Table 11. Broker IPOs and Analyst Forecast Quality (Monitoring Effects)

Dependent variable	(1) Bias	(2) Error	(3) Bias	(4) Error	(5) Bias	(6) Error
$IPOTreat[-1, +1]_{Star}$	-0.074 (-1.46)	-0.104** (-2.38)				
$IPOTreat[-1, +1]_{NonStar}$	-0.047 (-1.56)	-0.045 (-1.43)				
$IPOTreat[-1, +1]_{High_B_Exp}$			-0.069** (-2.21)	-0.060* (-1.93)		
$IPOTreat[-1, +1]_{Low_B_Exp}$			-0.058* (-1.69)	-0.047 (-1.38)		
$IPOTreat[-1, +1]_{InvestmentBank}$					-0.068*** (-2.78)	-0.062*** (-2.60)
$IPOTreat[-1, +1]_{PureResearch}$					-0.020 (-0.24)	0.028 (0.30)
$BrokerSize$	-0.009 (-1.63)	-0.016*** (-2.87)	-0.009 (-1.64)	-0.016*** (-2.87)	-0.009 (-1.63)	-0.016*** (-2.87)
$Horizon$	0.378*** (37.56)	0.547*** (54.77)	0.378*** (37.57)	0.547*** (54.78)	0.378*** (37.56)	0.547*** (54.78)
$Coverage$	0.015*** (2.91)	0.011* (1.94)	0.015*** (2.91)	0.011* (1.93)	0.015*** (2.92)	0.011* (1.94)
$Experience$	-0.016*** (-4.19)	-0.013*** (-3.31)	-0.016*** (-4.19)	-0.013*** (-3.31)	-0.016*** (-4.19)	-0.013*** (-3.31)
Analyst Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm-year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N	794,932	794,932	794,932	794,932	794,932	794,932
Adjusted R^2	0.797	0.875	0.797	0.875	0.797	0.875
Within Reg F-tests						
Diff	-0.027 (0.21)	-0.059 (1.21)	-0.011 (0.06)	-0.013 (0.08)	-0.048 (0.33)	-0.090 (0.89)

Notes. This table presents results from estimating a modified version of the generalized difference-in-differences Equation (8), where treatment brokers and analysts are partitioned into various groups. T-statistics are reported in parentheses, and standard errors are clustered by firm-year. All p -values are two tailed.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

analysts to produce higher-quality forecasts and that employee composition is affected as higher-quality employees are encouraged to stay throughout the IPO process.

Overall, our study provides new evidence on the impact of IPOs on the quality of analyst forecasts. Prior research examines the relation between forecast quality and the investment banking relationships of the brokerage, but we are unaware of any studies that explore the influences of the brokerage itself going public on forecast quality. Our study extends this literature by demonstrating that the brokerage house's own IPO can have a significant effect on the quality of the forecasts produced by the equity research division.

Acknowledgments

The authors thank John Hand, Melissa Lewis-Western, James Myers, Katherine Schipper (FARS discussant), Jake Thornock, and workshop participants at the 2019 FARS Editors' Picks Plenary Session, HKUST, University of California Irvine, University of North Carolina Chapel Hill, London Business School, University of Tennessee, Brigham Young University, Idaho State University, Singapore Management University, the 2021 McGill Accounting Research Conference, and the 2018 University of British Columbia, Oregon, and Washington Conference for helpful comments and suggestions. Brady Twedt's current affiliation is James Benjamin Department of Accounting at Mays Business School, Texas A&M University, College Station, Texas 77843.

Appendix. Variable Definitions

Variable	Definition
<i>All-star</i>	An indicator variable equal to one if the analyst is an All-star in year $t - 1$ and zero otherwise
<i>Avg_Bias</i>	Average value of <i>Bias</i> of all forecasts issued by an analyst in year $t - 2$
<i>Avg_Error</i>	Average value of <i>Error</i> of all forecasts issued by an analyst in year $t - 2$
<i>B_Exp</i>	The number of years the analyst has issued a forecast while employed by the IPO brokerage, as of year $t - 1$
<i>Bias</i>	The value of the forecast minus reported earnings, scaled by price as of two days prior to the forecast announcement date, and multiplied by 100
<i>BrokerSize</i>	The natural log of the number of analysts in the analyst's brokerage house who issued a forecast during the prior year
<i>Coverage</i>	The natural log of the number of firms the analyst forecasted for during the prior year
<i>Error</i>	The absolute value of <i>Bias</i>
<i>Experience</i>	The natural log of the number of years the analyst has issued a forecast for the firm of interest as of year $t - 1$
<i>Horizon</i>	The natural log of the number of days between the forecast and the fiscal period end date
<i>InvestmentBank</i>	An indicator variable equal to one if the brokerage is classified as a full service investment bank and zero otherwise
<i>IPOTreat[-1, +1]</i>	An indicator variable equal to one for forecasts issued by IPO analysts during the two-year period surrounding their brokerage's IPO and zero otherwise
<i>Num_Analysts</i>	The natural log of the number of analysts employed at the brokerage during year t
<i>Num_Firms</i>	The natural log of the total number of firms that the brokerage covers during year t
<i>Num_Forecasts</i>	The natural log of the total number of one-year ahead earnings forecasts that the brokerage issues during year t
<i>PostTreat[+2, +3]</i>	An indicator variable equal to one for forecasts issued by IPO analysts during the two-year period starting one year after the IPO date and zero otherwise
<i>PostTreat[4+]</i>	An indicator variable equal to one for forecasts issued by IPO analysts during the period starting four years after the IPO date and zero otherwise
<i>PreTreat[-3, -2]</i>	An indicator variable equal to one for forecasts issued by IPO analysts during the two-year period ending one year before the IPO date and zero otherwise
<i>PureResearch</i>	An indicator variable equal to one if the brokerage is classified as a research only brokerage and zero otherwise
<i>RecRevision</i>	The difference between an analyst's current recommendation and prior recommendation for the covered firm, on a five-point scale
<i>Stay</i>	An indicator variable equal to one if the analyst remains at the brokerage during the entire two-year IPO window and zero otherwise

Endnotes

¹ See, for example, <https://news.bloomberglaw.com/bloomberg-law-analysis/analysis-three-decades-of-ipo-deals-1990-2019>.

² For a review of this literature, see Ramnath et al. (2008).

³ For example, Falato and Scharfstein (2016) provide evidence that post-IPO pressure to maximize short-term stock price performance increases risk-taking by bank IPOs.

⁴ We focus on a two-year window because pre-IPO planning can take 12–24 months (PwC 2017). In later analyses, we expand this window to longer horizons to assess whether IPOs have lasting effects on analysts' forecast quality.

⁵ In supplementary analyses, we also examine stock recommendation revision optimism bias as an alternative measure of bias.

⁶ See Table 1 for details on our sample of IPOs. Our initial search for IPOs on ThomsonOne was for 1985–2014, but the latest broker IPO occurred in 2007.

⁷ Our findings are also robust to the use of an alternative, relative, forecast quality measure (Clement 1999) instead of including fixed effects.

⁸ We also demonstrate that our results are robust to alternative standard error clustering methods, including covered-firm and year, broker and year, and analyst and year.

⁹ For an extensive review of this literature, see Mehran and Stulz (2007). Although there is some mixed evidence on the impact of investment banking incentives, most studies suggest that such activities reduce analysts' objectivity.

¹⁰ Despite the regulatory initiative, some studies suggest that conflicts of interest remain in the sell-side analyst industry (see, e.g., Mayew 2008, Corwin et al. 2017, Horton et al. 2017, Dambra et al. 2018).

¹¹ We begin our sample of IPOs in 1985 because the I/B/E/S data set begins in 1982, and we require pre-IPO forecast data for our tests. The I/B/E/S earnings forecast database is less populated in the 1980s. Our results are robust to partitioning on later years (i.e., years after 1990). Additionally, although we searched for IPOs through 2014, the last IPO meeting all of our sample inclusion criteria occurred in 2007 (see Table 1).

¹² We remove these observations to increase the power of our analyses examining pre- versus post-IPO effects. There are 1,362 forecasts during this blackout period, representing only 0.17 percent of our sample; in untabulated analyses, we find consistent results if we instead retain these forecasts.

¹³ In Table 4, we also control for additional measures reflecting broker resources, including number of forecasts and number of firms covered. Our inferences remain unchanged.

¹⁴ We note that the mean level of firm-specific experience in our sample is one to two years lower than that observed in prior studies. This is likely because we measure experience as of year $t-1$, as opposed to year t .

¹⁵ We cannot consider broker $\times$ year fixed effects because our treatment effect varies at the broker $\times$ year level.

¹⁶ In untabulated analysis, we also examine quarterly earnings forecasts using either one-, two-, three-, or four-quarter-ahead forecasts as forecast quality alternatively measured and price targets as an alternative measure of analyst bias. The results provide no evidence of significant changes in bias of quarterly forecasts or price targets around IPOs. Additionally, we examine market reactions to forecast revisions as an indirect measure of analyst forecast quality. We find some evidence of stronger market reactions to IPO analysts, but the results are not robust to additional controls for contemporaneous earnings announcements or earnings surprise.

¹⁷ We classify brokerages into full-service investment banks and pure research firms by reading the company description provided

in the prospectus. When we could not locate a prospectus, we used the company description of the first 10-K filed in EDGAR. We classify that 7 of the 23 IPO brokerage as pure research, including Conning Corp, Southwest Securities Group, TD Waterhouse Group, The Charles Schwab Corp, Instinet Group, Ragen MacKenzie Group, and Oppenheimer Capital.

References

- Badertscher B, Jorgensen B, Katz S, Kinney W (2014) Public equity and audit pricing in the United States. *J. Accounting Res.* 52(2): 303–339.
- Ball R, Shivakumar L (2008) Earnings quality at initial public offerings. *J. Accounting Econom.* 45(2–3):324–349.
- Becker C, Defond M, Jiambalvo J, Subramanyam K (1998) The effect of audit quality on earnings management. *Contemporary Accounting Res.* 15(1):1–24.
- Bouchet C, Guillemin S, Briancon S (1996) Nonspecific effects in longitudinal studies: Impact on quality of life measures. *J. Clinical Epidemiology* 49(1):15–20.
- Bourveau T, Lou Y, Wang R (2018) Shareholder litigation and corporate disclosure: Evidence from derivative lawsuits. *J. Accounting Res.* 56(3):797–842.
- Bradshaw MT, Richardson SA, Sloan RG (2006) The relation between corporate financing activities, analysts' forecasts and stock returns. *J. Accounting Econom.* 42(1–2):53–85.
- Brown LD, Call AC, Clement MB, Sharp NY (2019) Managing the narrative: Investor relations officers and corporate disclosure. *J. Accounting Econom.* 67(1):58–79.
- Cecchini M, Jackson S, Liu X (2012) Do initial public offering firms manage accruals? Evidence from individual accounts. *Rev. Accounting Stud.* 17(1):22–40.
- Christensen HB, Hail L, Leuz C (2016) Capital-market effects of securities regulation: Prior conditions, implementation, and enforcement. *Rev. Financial Stud.* 29(11):2885–2924.
- Chung R, Firth M, Kim J (2002) Institutional monitoring and opportunistic earnings management. *J. Corporate Finance* 8(1):29–48.
- Clement M (1999) Analyst forecast accuracy: Do ability, resources, and portfolio complexity matter? *J. Accounting Econom.* 27(3):285–303.
- Clement MB, Tse SY (2003) Do investors respond to analysts' forecast revisions as if forecast accuracy is all that matters? *Accounting Rev.* 78(1):227–249.
- Corwin S, Larocque S, Stegemoller MA (2017) Investment banking relationships and analyst affiliation bias: The impact of the global settlement on sanctioned and non-sanctioned banks. *J. Financial Econom.* 124(3):614–631.
- Cowen A, Groyberg B, Healy P (2006) Which types of analyst firms are more optimistic? *J. Accounting Econom.* 41(1–2):119–146.
- Dambra M, Field L, Gustafson M, Pisciotta K (2018) The consequences to analyst involvement in the IPO process: Evidence surrounding the JOBS Act. *J. Accounting Econom.* 65(2):302–330.
- Dechow PM, Hutton AP, Sloan RG (2000) The relation between analysts' forecasts of long-term earnings growth and stock price performance following equity offerings. *Contemporary Accounting Res.* 17(1):1–32.
- Dugar A, Nathan S (1995) The effect of investment banking relationships on financial analysts' earnings forecasts and investment recommendations. *Contemporary Accounting Res.* 12(1):131–160.
- Dyck A, Volchkova N, Zingales L (2008) The corporate governance role of the media: Evidence from Russia. *J. Finance* 63(3):1093–1135.
- Ellingsen T, Kristiansen EG (2022) Fair and square: A retention model of managerial compensation. *Management Sci.* 68(5):3604–3624.
- Ertimur Y, Sletten E, Sunder J, Weber J (2018) When and why do IPO firms manage earnings? *Rev. Accounting Stud.* 23(3):872–906.
- Falato A, Scharfstein D (2016) The stock market and bank risk-taking. NBER Working Paper No. 22689, National Bureau of Economic Research, Cambridge, MA.

- Fang L, Yasuda A (2009) The effectiveness of reputation as a disciplinary mechanism in sell-side research. *Rev. Financial Stud.* 22(9): 3735–3777.
- Gu Z, Wu JS (2003) Earnings skewness and analyst forecast bias. *J. Accounting Econom.* 35(1):5–29.
- Gu Z, Li Z, Yang YG (2013) Monitors or predators: The influence of institutional investors on sell-side analysts. *Accounting Rev.* 88(1):137–169.
- Guiso L, Sapienza P, Zingales L (2015) The value of corporate culture. *J. Financial Econom.* 117(1):60–76.
- Heaton J (2002) Managerial optimism and corporate finance. *Financial Management* 31(2):33–45.
- Hong H, Kacperczyk M (2010) Competition and bias. *Quart. J. Econom.* 125(4):1683–1725.
- Horton J, Serafeim G, Wu S (2017) Career concerns of banking analysts. *J. Accounting Econom.* 63(2):231–252.
- Irvine PJ (2004) Analysts' forecasts and brokerage-firm trading. *Accounting Rev.* 79(1):125–149.
- Jackson A (2005) Trade generation, reputation, and sell-side analysts. *J. Finance* 60(2):673–717.
- Jacob J, Lys TZ, Neale MA (1999) Expertise in forecasting performance of security analysts. *J. Accounting Econom.* 28(1): 51–82.
- Jacob J, Rock S, Weber DP (2008) Do non-investment bank analysts make better earnings forecasts? *J. Accounting Auditing Finance* 23(1):23–61.
- Kadan O, Madureira L, Wang R, Zach T (2009) Conflicts of interest and stock recommendations: The effects of the Global Settlement and related regulations. *Rev. Financial Stud.* 22(10):4189–4217.
- Kedia S, Rajgopal S (2011) Do the SEC's enforcement preferences affect corporate misconduct? *J. Accounting Econom.* 51(3):259–278.
- Kedia S, Rajgopal S, Zhou X (2014) Did going public impair Moody's credit ratings? *J. Financial Econom.* 114(2):293–315.
- Kole SR (1997) The complexity of compensation contracts. *J. Financial Econom.* 43(1):79–104.
- Lin H, McNichols M (1998) Underwriting relationships, analysts' earnings forecasts and investment recommendations. *J. Accounting Econom.* 25(1):101–127.
- Ljungqvist A, Marston FC, Wilhelm WJ (2006) Competing for securities underwriting mandates: Banking relationships and analyst recommendations. *J. Finance* 61(1):301–340.
- Lowry M, Shu S (2002) Litigation risk and IPO underpricing. *J. Financial Econom.* 65:309–335.
- Mayew WJ (2008) Evidence of management discrimination among analysts during earnings conference calls. *J. Accounting Res.* 46(3): 627–659.
- Mehran H, Stulz R (2007) The economics of conflicts of interest in financial institutions. *J. Financial Econom.* 85(2):267–296.
- Merkley K, Michael R, Pacelli J (2017) Does the scope of sell-side analyst industry matter? An examination of bias, accuracy and information content of analyst reports. *J. Finance* 72(3):1285–1334.
- Michael R, Womack K (1999) Conflict of interest and the credibility of underwriter analyst recommendations. *Rev. Financial Stud.* 12(4):653–686.
- O'Brien PC, McNichols MF, Hsiou-Wei L (2005) Analyst impartiality and investment banking relationships. *J. Accounting Res.* 43(4):623–650.
- Oyer P, Schaefer S (2005) Why do some firms give stock options to all employees? An empirical examination of alternative theories. *J. Financial Econom.* 76(1):99–133.
- Pacelli J (2019) Corporate culture and analyst catering. *J. Accounting Econom.* 67(1):120–143.
- PwC (2017) Roadmap for an IPO: A guide to going public. Accessed https://www.pwc.com/hu/hu/szolgaltatasok/konyvvizsgalat/szamviteli-tanacsadas/kiadvanyok/roadmap_for_an_ipo.pdf.
- Ramnath S, Rock S, Shane P (2008) The financial analyst forecasting literature: A taxonomy with suggestions for further research. *Internat. J. Forecasting* 24(1):34–75.
- Ritter J, Welch I (2002) A review of IPO activity, pricing, and allocations. *J. Finance* 57(4):1795–1828.
- Sweeney A (1994) Debt-covenant violations and managers' accounting responses. *J. Accounting Econom.* 17(3):281–308.
- Teoh S, Welch I, Wong T (1998) Earnings management and the long-run market performance of initial public offerings. *J. Finance* 53(6):1935–1974.
- Yu F (2008) Analyst coverage and earnings management. *J. Financial Econom.* 88(2):245–271.